Predictive Targeting and Segmentation of Food-Insecure Households

EE 5290 Final Project · Spring 2026

Iowa State University · May 2026

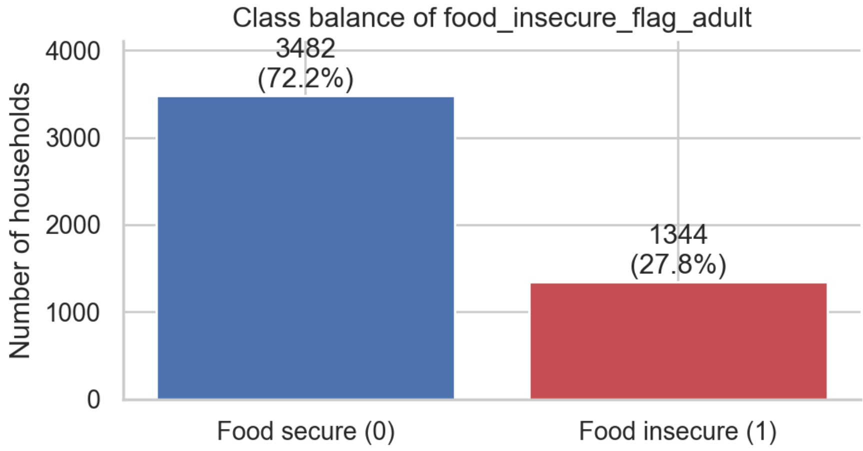
The Problem

Food-assistance demand is rising while organisational capacity is fixed. Food banks must decide **which households to prioritize** and **what programs to offer**.

Two questions:

- **Prediction:** Can we identify food-insecure households accurately enough to support triage?
- **Segmentation:** What *kinds* of food-insecure households exist, and how should that shape outreach?

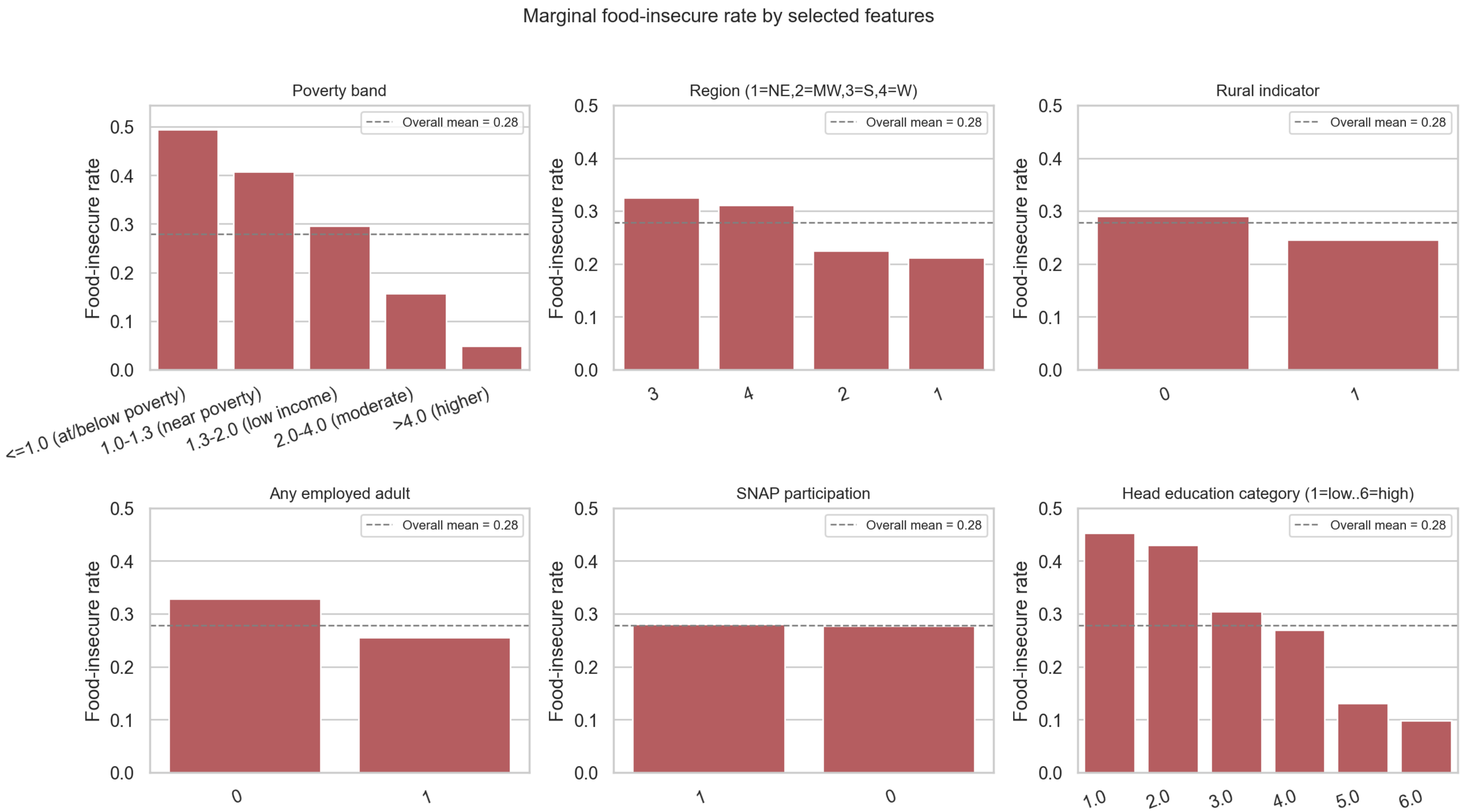
Approach: a two-track ML analysis joined by a synthesis section.



Data

- Provided FoodAPS extract: **4,826 households**, 38 columns
- Outcome: food_insecure_flag_adult (binary)
- **27.85% positive** — moderate imbalance
- Sentinel codes (-996/-997/-998) recoded; caraccess dropped (92% missing)
- Distance variables logged; one-hot encoding for nominals
- Stratified 80/20 split; 5-fold CV on train only

EDA: Where the Risk Lives



Poverty band, employment, SNAP, and education move the food-insecurity rate by 20-30 percentage points each. Rural households are slightly elevated.

Track 1 — Predictive Modelling Setup

- Three classifiers: **Logistic Regression**, **SVM (RBF)**, **Random Forest**
- Three imbalance strategies: **none**, **balanced class weights**, **F1-tuned threshold**
- Tuning metric: **average precision (PR-AUC)**
- Threshold tuned on training out-of-fold predictions (test set untouched)

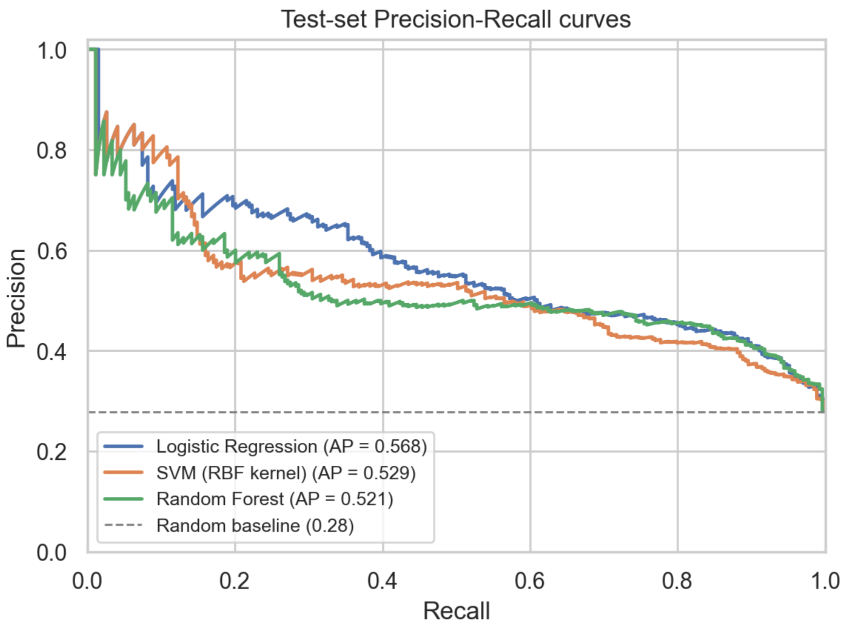
Why PR-AUC? With 28% positives, accuracy is misleading (72% trivially). PR-AUC ignores the abundant true negatives.

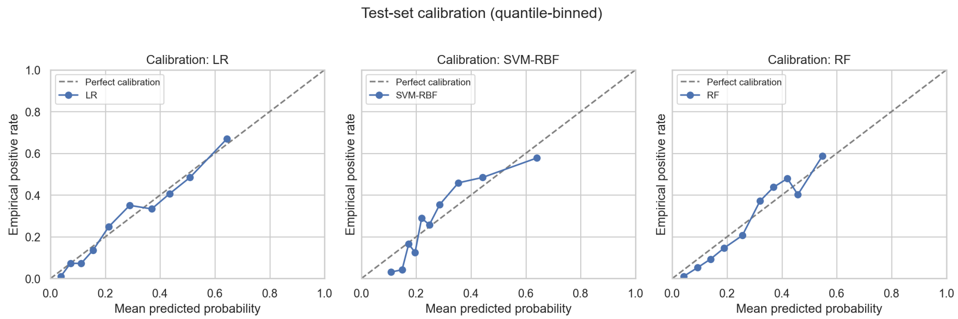
Track 1 — Test Set Results (9 cells)

Model (strategy)	thr	PR-AUC	Prec.	Rec.	F1
LR -- no adjust	0.50	0.568	0.624	0.364	0.460
LR -- balanced	0.50	0.568	0.463	0.762	0.576
LR -- F1-tuned thr	0.289	0.568	0.466	0.755	0.576
SVM-RBF -- no adjust	0.50	0.529	0.549	0.208	0.302
SVM-RBF -- balanced	0.50	0.527	0.564	0.346	0.429
SVM-RBF -- F1-tuned thr	0.236	0.529	0.425	0.747	0.542
RF -- no adjust	0.50	0.521	0.602	0.186	0.284
RF -- balanced	0.50	0.510	0.485	0.665	0.561
RF -- F1-tuned thr	0.301	0.521	0.453	0.792	0.577

LR wins on every aggregate metric. Threshold tuning ≈ class weights. The choice of *threshold* matters more than the choice of *weighting*.

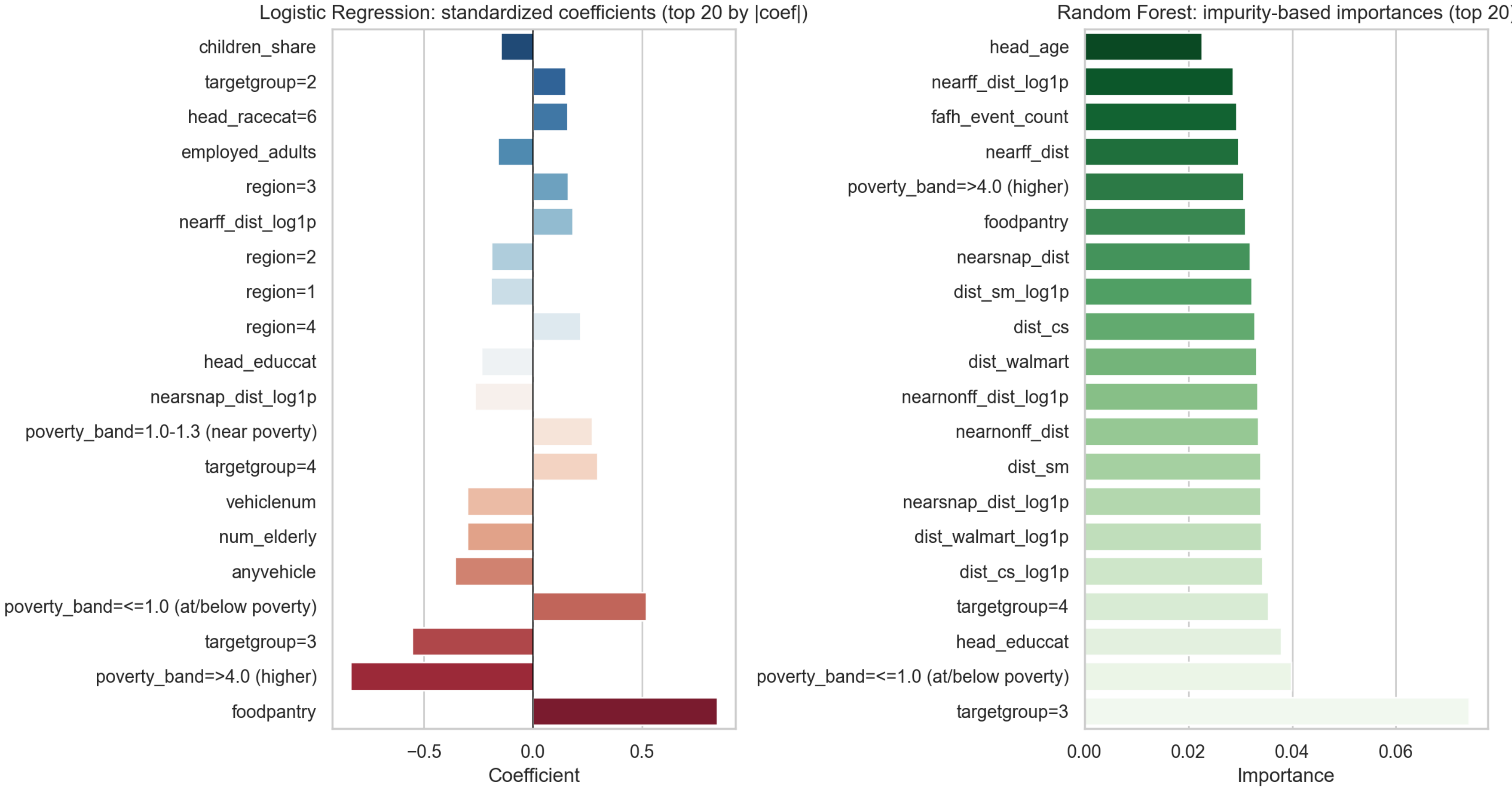
Track 1 — PR Curves and Calibration





Left: LR dominates SVM and RF in the high-recall regime that matters for triage. Right: LR is well calibrated; RF mildly overconfident at the extremes.

Track 1 — What the Models Look At



LR (left, signed): poverty band, food-pantry use, vehicles, education, distances. RF (right, unsigned) emphasises the same features plus the cluster of distance variables.

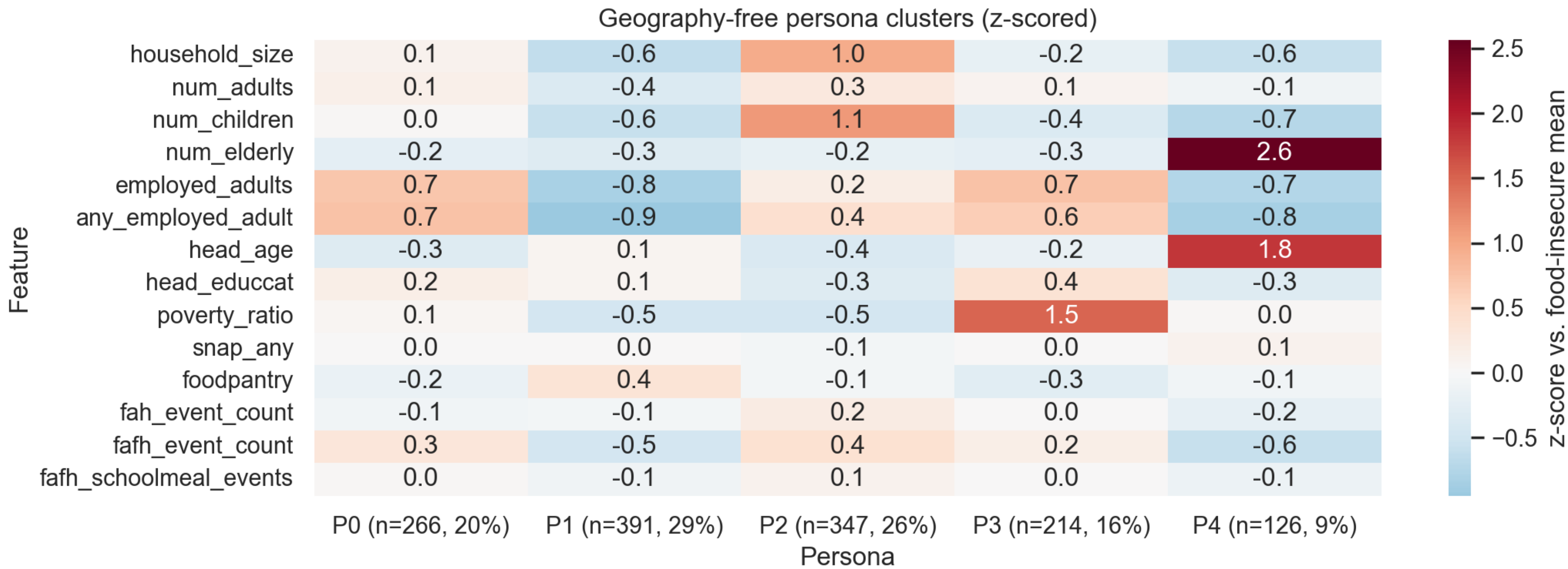
Track 2 — Segmentation Setup

- Filter to the 1,344 food-insecure households
- Standardize → PCA (24 PCs explain 80% of variance)
- K-Means sweep k=2..8; choose by silhouette + elbow
- Stability check across 8 random initializations (mean ARI > 0.9)

Two views:

- **Geographic** (all features): silhouette winner is **k=2**, a strong rural / urban access split.
- **Geography-free** (drop distances/region): exposes **5 personas** that differ on composition, employment, programs.

Track 2 — Five Personas Among the Food Insecure

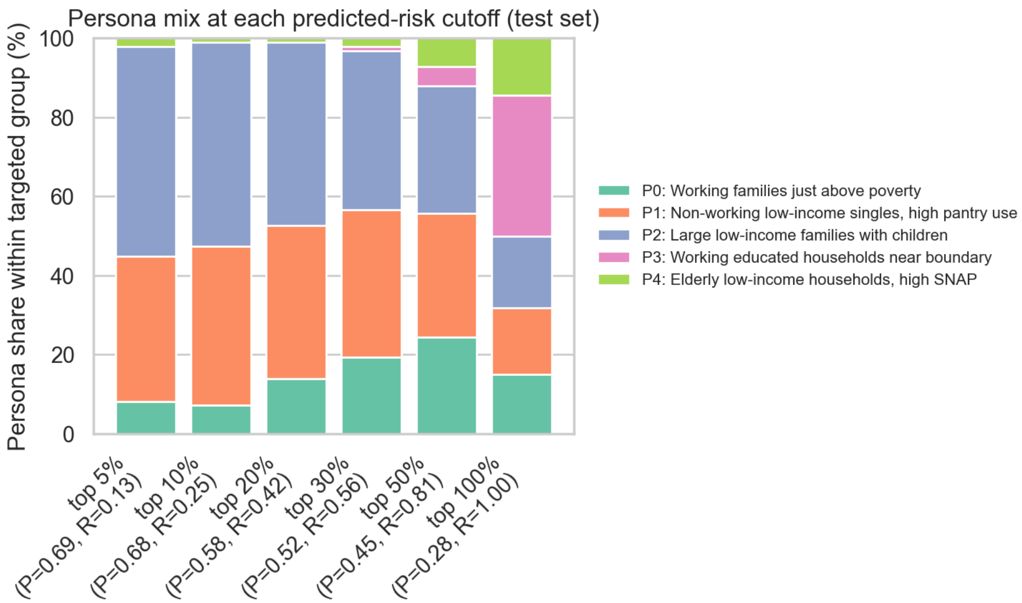


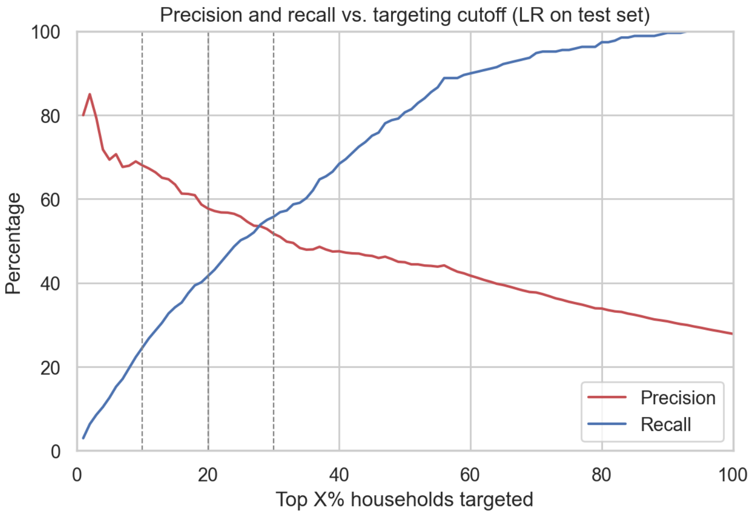
Z-scores within the food-insecure subset. Red = above the food-insecure mean.

Track 2 — The Five Personas

P	Persona	n	Share	Headline feature
P0	Working families just above poverty	266	20%	100% employed; pov. ratio 1.43
P1	Non-working low-income singles	391	29%	Pantry use 28% (3x avg); 16% emp.
P2	Large low-income families with children	347	26%	Mean household 5.0; 2.6 kids; pov. 0.85
P3	Working educated, near boundary	214	16%	Pov. ratio 3.05 (above poverty!)
P4	Elderly low-income, high SNAP	126	9%	Mean head age 70; 42% on SNAP

Synthesis — Who Does the Model Flag?





Top 10% predicted-risk: 88% of flags are P1 or P2. P3 (working educated) and P4 (elderly on SNAP) get <2% of flags.

Targeting Recommendations

- **Top decile:** precision 67%, recall 24% → 97 of 966 households; 66 truly food-insecure
- **P1 (45%) — non-working low-income singles:** pantry partnerships, supplemental distribution
- **P2 (43%) — large families with children:** school-meal coordination, bulk-pack programmes
- **P0 (12%) — working families just above poverty:** light-touch SNAP-enrolment outreach
- **P4 — elderly on SNAP:** monitor via existing partnerships, do not disengage on low scores
- **P3 — working educated:** qualitative follow-up; not reachable by income-eligibility programmes

All findings are associations, not causal claims. Survey weights and external county-level data are obvious next steps.

Thank you. Questions?

Code & figures: notebooks/, src/, report/

Reproduction: python -m venv venv && pip install -r requirements.txt